

FEB12001X – Midterm 17 September 2025

FEB12001X Applied Microeconomics 25/26 (ESE) · 4 exercises · 10.0 points

Exercise 1 Public Economics #50860151

2 pts · Last updated 19 Sep, 2025, 10:41

TEXT BLOCK

Consider two individuals, A and B, who both derive utility from consuming a pure public good. For individual A, the utility from consuming Q units of the public good (measured in euros) is given by

$$U^A = aQ - \frac{1}{2} Q^2$$

The utility of individual B from consuming Q units of the public good (measured in euros) is given by

$$U^B = bQ - \frac{1}{2} Q^2$$

The marginal production cost of the public good are equal to c euro, which is also the price per unit of the public good on the market. Assume that $a > b > c > 0$.

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Determine the socially efficient level of provision of the public good

Either by maximizing $U^A + U^B - cQ$

1 pt

Either by maximizing $U^A + U^B - cQ$ or by using that for efficient provision of the public good it must be that $MU^A + MU^B = MC$, we get $a - Q + b - Q - c = 0$, such that $Q = (a + b - c)/2$.

1 PT · OPEN · 1/2 PAGE

Suppose A and B decide independently how many units of the public good to buy (of course taking the presence of the other into account). Determine how many units person B consumes in the Nash equilibrium.

In the Nash equilibrium, person A provides $Q = a - c$ units. Person B provides 0 units. As the public good is non-rival, both consume $a - c$ units of the public good.

1 pt

In the Nash equilibrium, person A provides $Q = a - c$ units. Person B provides 0 units. As the public good is non-rival, both consume $a - c$ units of the public good.

(Note that the above is indeed a Nash equilibrium: if B provides nothing, the "best response" of A is to maximise own utility by setting $MU^A = c$, which yields $Q = a - c$. At $Q = a - c$, the marginal benefits of consumption of person B (equal to $b - Q$) is below marginal cost, as $b - (a - c) < c$ as it is assumed that $a > b$).

Correct Nash equilibrium, but incorrect conclusion regarding consumption of B.

0.5 pts

Correct Nash equilibrium, but incorrect conclusion regarding consumption of B.

Exercise 2 Public Economics #50860152

3 pts · Last updated 22 Sep, 2025, 09:33

TEXT BLOCK

Suppose that person A produces good x. The production process imposes external cost on person B. Both persons maximise their own utility, which increases in their own income. The profit that person A derives from producing q units of good x is given by $kq - \frac{1}{2}q^2$ euro. Total external cost incurred by person B are equal to cq euro. The conditions of the Coase Theorem are met and Person A has the (property) right to produce x. In the absence of government intervention, Person A and B make a socially efficient binding agreement, which determines how many units of good x person A can produce as well as a transfer of T euro from person B to person A. Assume that $k > c > 0$.

1 PT · OPEN · 3/5 PAGE

Show that the minimal transfer T that person B pays to person A equals $\frac{1}{2}c^2$.

Coase Theorem implies that ...

1 pt

Coase Theorem implies that person A produces the socially efficient level of q rather than the profit-maximising level of q . Person A only accepts to reduce production if the resulting reduction in profit is compensated for by the transfer T. The privately-optimal q is given by maximizing profit with respect to q . This gives first-order condition $k - q = 0$, so that $q = k$. This yields profit $\frac{1}{2}k^2$. Socially efficient level of q is given by maximizing profit minus external cost. This yields first-order condition $k - q - c = 0$, so that $q^* = k - c$. This yields profit $k^*(k-c) - \frac{1}{2}(k-c)^2$. Hence, the minimal $T = \frac{1}{2}k^2 - [k^*(k-c) - \frac{1}{2}(k-c)^2] = \frac{1}{2}c^2$. The final step can also be done (implicitly via a graph) as $(1/2)*[k - (k-c)]*c$ or using an integral.

1 PT · OPEN · 2/5 PAGE

Explain why the minimal transfer T increases in marginal external cost c . Use no more than 3 sentences.

A higher c implies that the...

1 pt

A higher c implies that the socially efficient level of x is lower (more restricted). Hence, the agreement induces person A to reduce production of x more (the difference between the privately optimal and the socially efficient level of x is larger if c is larger). A larger reduction in x implies a larger loss in utility (or profit) for person A, which must be compensated for by the transfer. Note: Answers like 'the transfer is higher because the external cost are higher' or 'if c is higher, person B is willing to pay more to reduce x .' receive 0 points (these are not wrong statements, but they do not explain why the *minimal* transfer increases in c).

Correct reasoning that high...

0.5 pts

Correct reasoning that higher c implies that the socially efficient level of x is lower, but no link to utility / profit of person A.

1 PT · OPEN · 1/4 PAGE

Suppose the government maximises a social welfare function, which is equal to the sum of the utilities of persons A and B. Give one reason for why the government may decide to engage in government intervention in the situation of this exercise. Use no more than one sentence.

For distributional reasons ...

1 pt

For distributional reasons / because the government prefers to redistribute from A to B / because marginal utilities of income are different.

Note: "to improve efficiency / correct market failure is wrong", as efficiency is already maximal (Coase Theorem).

Exercise 3 Personnel economics #51095228

5 pts · Last updated 4 Sep, 2025, 10:44

TEXT BLOCK

A profit-maximizing tobacco company hires a worker. The worker's utility function is:

$U = w - \frac{1}{2}\theta e^2 - \beta q$, where w is the worker's income, e is the amount of effort exerted by the worker, and q is the amount of output produced by the worker. θ and β are positive parameters. The worker's income consists of a bonus b per unit of production. There is no base salary. Hence, $w = bq$. The production function is $q = \mu e$. The first term of the utility function reflects that the worker enjoys income. The second term reflects worker's cost of effort. And the third term reflects that the worker feels guilt about producing tobacco, which reduces his utility. The worker chooses to work for the tobacco company if his utility is at least equal to zero. Effort e cannot be negative.

2 PTS · OPEN · 1/2 PAGE

Derive the level of effort the worker chooses to exert.

Substitute the income equat...

1 pt

Substitute the income equation and production function into the utility function:

$U = b\mu e - \frac{1}{2}\theta e^2 - \beta\mu e$. To find optimal effort, differentiate and set equal to zero:
 $b\mu - \theta e - \beta\mu = 0$.

Solve for e , which resu...

0.5 pts

Solve for e , which results in $e = \frac{\mu}{\theta}(b - \beta)$.

Hence, since effort cannot ...

0.5 pts

Hence, since effort cannot be negative, effort is zero when $b < \beta$

3 PTS · OPEN · 4/5 PAGE

Assume each unit of production is sold by the tobacco company for price p , which is assumed sufficiently high so that the tobacco company wishes to produce a positive amount. Derive the profit-maximizing level of the bonus b .

Profits equal $pq - bq$.

1 pt

Profits equal $pq - bq$.

Substitute the level of eff...

1 pt

Substitute the level of effort chosen by the worker as derived under part 3a. This results in:

$$(p\mu - b\mu) \left(\frac{\mu}{\theta}(b - \beta) \right)$$

Take the derivative to the ...

1 pt

Take the derivative to the bonus and set equal to zero. Solving gives: $b = \frac{p + \beta}{2}$

Additional answer space #50629709

0 pts · Last updated 4 Sep, 2025, 15:30

TEXT BLOCK

Only use this extra space if you made a mistake in one of your answers. Please do not forget to **indicate in the answer box of that particular question where the correct answer can be found.**

Otherwise, the answers in the additional space below will not be reviewed.

NB: Do **not** use this additional space for answers that are too long.

OPEN · 4/5 PAGE

Additional answer box 1

OPEN · 1/2 PAGE

Additional answer box 2

OPEN · 1/2 PAGE

Additional answer box 3